

BEGINNER WORKSHOP – TRACK 1

NLP Fundamentals

| Understand how machines process, interpret, and generate human language.

HOSTED BY

Nawel

DURATION

1 Hour

DEFINITION

What is NLP?

Natural Language Processing is a field of AI that enables computers to understand, interpret, and generate human language.

THE CORE CHALLENGE

Human: "I love this movie" → Obvious emotion.

Machine: Binary character sequence without meaning.

Solution: Transform text into numerical data.

APPLICATIONS

- Chatbots & Assistants
- Sentiment Analysis
- Machine Translation
- Spam Filters

The Classic NLP Pipeline

01

COLLECTION

Gathering raw text from the web, files, or databases.

02

PREPROCESSING

Cleaning, normalization, and noise removal.

03

REPRESENTATION

Converting text into usable numerical vectors.

04

TRAINING

Learning patterns with a Machine Learning model.

05

EVALUATION

Measuring performance and making adjustments.

"Output quality depends directly on the quality of the initial cleaning."

PREPROCESSING

Data Cleaning

Reducing "noise" and standardizing text so the model can focus on the core meaning.

KEY STEPS

- Lowercasing
- Removing Punctuation
- Removing Numbers
- Stopword Removal

CONCRETE EXAMPLE

ORIGINAL TEXT

"Cats are running quickly!"



AFTER CLEANING

"cat run quick"

The text is now ready for vectorization.

FOUNDATIONAL STEP

Tokenization

The process of splitting raw text into smaller, meaningful units called **tokens**. It is the first crucial step in any NLP pipeline.

SEGMENTATION EXAMPLE

| *"I love NLP"*

I love NLP

TYPES OF TOKENIZATION

WORD-BASED

Classic splitting based on spaces and punctuation.

CHARACTER-BASED

Fine-grained analysis, useful for specific languages or models.

SENTENCE-BASED

Useful for analyzing long and complex documents.

SUBWORD-BASED

Modern compromise used by models like BERT or GPT.

CONCEPT

Stopwords

Very common words that carry little semantic value for global analysis.

TYPICAL EXAMPLES

the

is

and

in

at

of

IMPACT OF FILTERING

By removing these words, we reduce data size and improve model efficiency by focusing on meaningful terms.

ORIGINAL TEXT

"The cat is on the table"

↓ Stopword Filtering

RESULT

cat table

Stemming vs Lemmatization

Stemming

Brute-force cutting of word endings to keep only the "root" (stem).

```
running → run  
runs → run  
studies → studi
```

- + Very fast and simple.
- Sometimes inaccurate (roots aren't always words).

Lemmatization

Uses a dictionary and grammar to find the canonical form (lemma).

```
better → good  
mice → mouse  
was, is, are → be
```

- + Highly accurate and semantic.
- Slower, requires a dictionary/lexicon.

THE PROBLEM

Text Representation

Mathematical algorithms don't read letters. They only understand **numbers** and **calculations**.

THE SOLUTION: VECTORIZATION

Converting each word or document into a numerical vector (a list of numbers) that captures its features.

[0.12, -0.45, 0.89, 0.03, ...]

CLASSIC TECHNIQUES

BAG OF WORDS (BOW)

Counts the frequency of word occurrences in the text.

TF-IDF

Weights word importance based on their rarity.

WORD EMBEDDINGS

Captures semantic meaning (e.g., Word2Vec, GloVe).

ONE-HOT ENCODING

Simple binary representation of each word in the vocabulary.

VECTORIZATION

Bag of Words Model

A global vocabulary is created, and the frequency of each word's occurrence in a document is counted.

EXAMPLE

S1: "I love NLP"
S2: "I love AI"

MATRIX REPRESENTATION

Document	I	love	NLP	AI
S1	1	1	1	0
S2	1	1	0	1

MAJOR LIMITATION: Word order is completely lost. "The cat eats the mouse" = "The mouse eats the cat".

IMPROVEMENT

TF-IDF: Weighting Importance

*"Not all words have the same value.
A rare word is often more
informative than a frequent one."*

THE MATHEMATICAL LOGIC

TF – TERM FREQUENCY

Measures how often a word appears in **a specific document**. The more it appears, the more important it is locally.

IDF – INVERSE DOCUMENT FREQUENCY

Penalizes words that appear in **too many documents** (e.g., "the", "is"). If it's everywhere, it doesn't help distinguish.

Result: Rare Words = High Weight | Common Words = Low Weight

This allows the model to focus on the terms that truly characterize the subject of a text.

PRE-TRANSFORMERS ERA

Classic Models

- Naive Bayes
- Logistic Regression
- SVM (Support Vector Machines)

TYPICAL TASKS

Text classification, spam detection, binary sentiment analysis.

THE TECHNOLOGICAL WALL

The Absence of Context

Classic approaches treat words as isolated entities. They ignore word order and complex relationships within a sentence.

POLYSEMY EXAMPLE

*"I'm going to the **bank** for a loan."*

vs

*"He sat on the river **bank**."*

The Bag-of-Words model sees the same word, but the meaning is radically different.

SUMMARY

Key Takeaways

- 01. Raw text must be **cleaned and normalized** before any analysis.
 - 02. **Tokenization** is the foundational step for segmenting language.
 - 03. **Vectorization** (BoW, TF-IDF) allows machines to "read" text.
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The Future of NLP

Modern models (Transformers like BERT and GPT) overcome classic limitations by understanding **global context** and semantic nuances.

Thank You!

QUESTIONS?